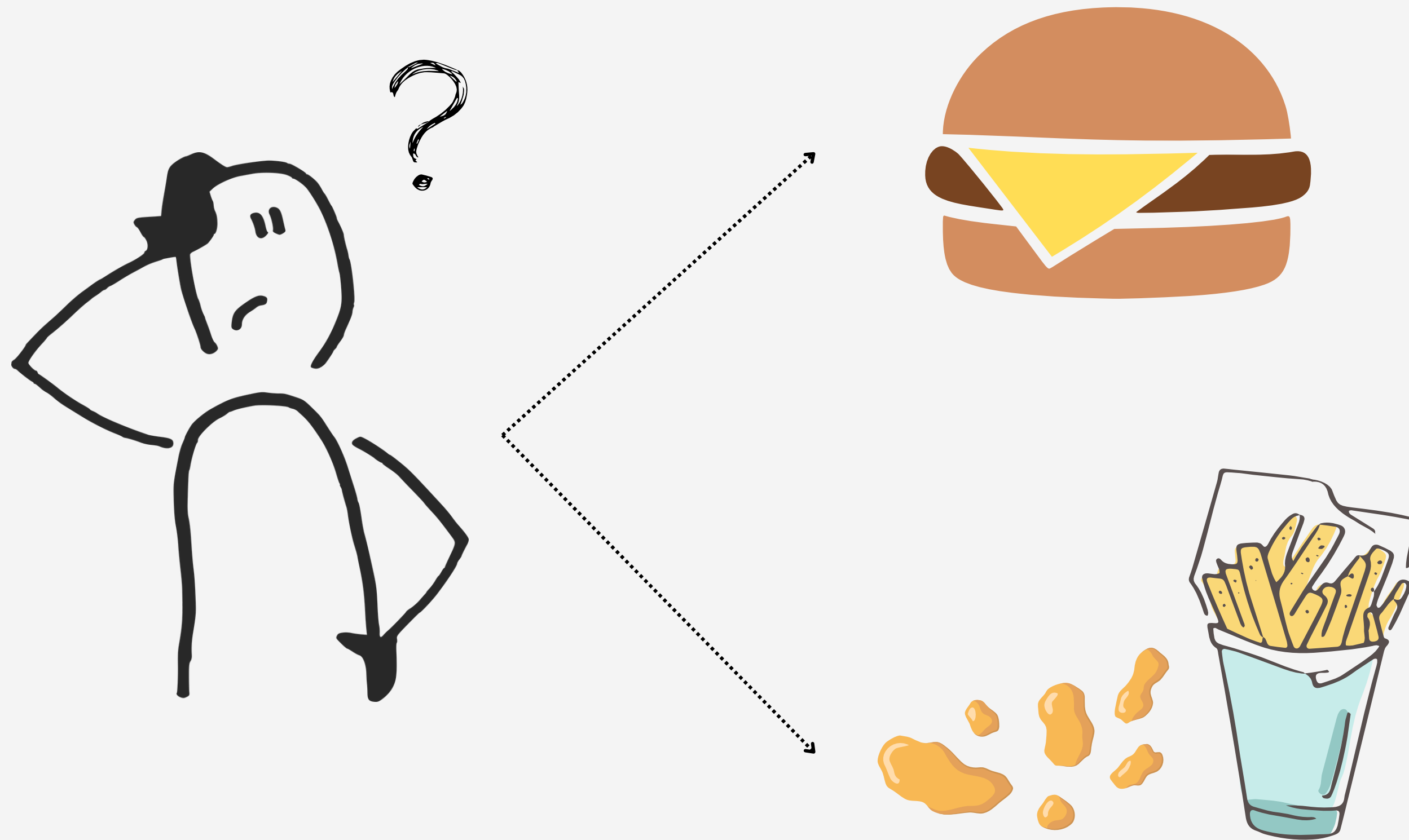


WHAT IF THAT C.ELEGANS COULD ACTUALLY THINK ?



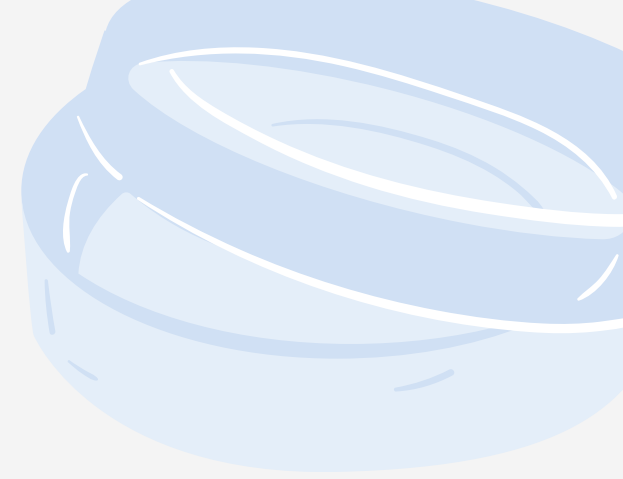


DO VERY SIMPLE ORGANISMS FOLLOW SIMILAR STRATEGIES WHEN THEY
MOVE BETWEEN FOOD PATCHES ?

BEHAVIORAL OPTIMIZATION: TESTING OPTIMAL FORAGING THEORY

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01 Introduction

- Biological question
- Theory

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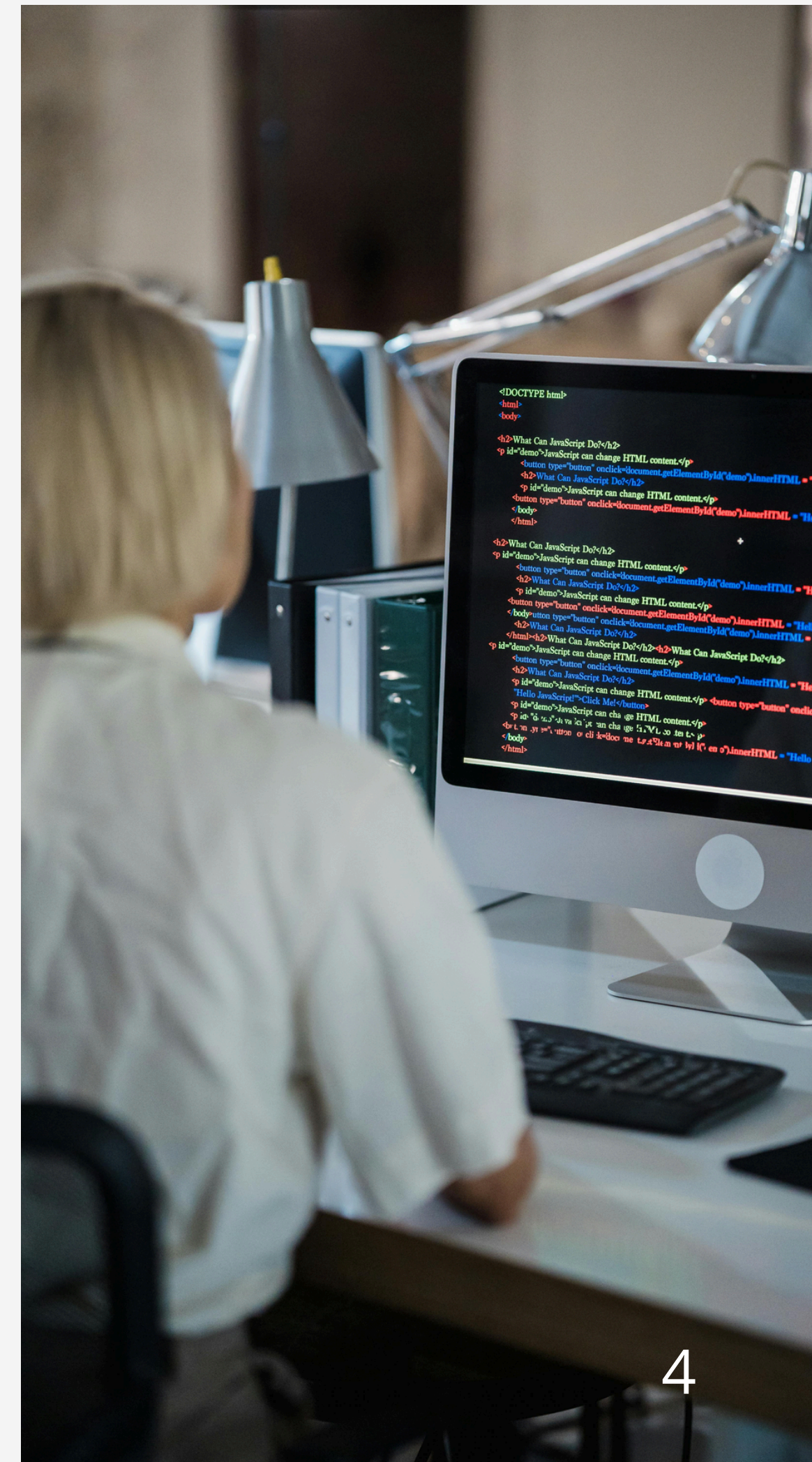
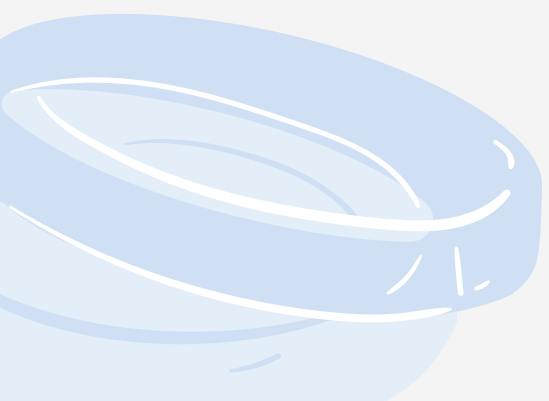
- Procedure
- Analysed the dataset

03 Results

- What we obtained ?

04 Discussion

- Interpretation



Introduction

- **The Problem:** Why do organisms leave a patch of food?
- **Significance:** Understanding the mathematics of efficiency in the context of *C.elegans* worms.



Methodology

Phase 01
Overview of the dataset



Phase 03
Analysis of Multiple Worms



Phase 02
Creation of a Clean Dataset

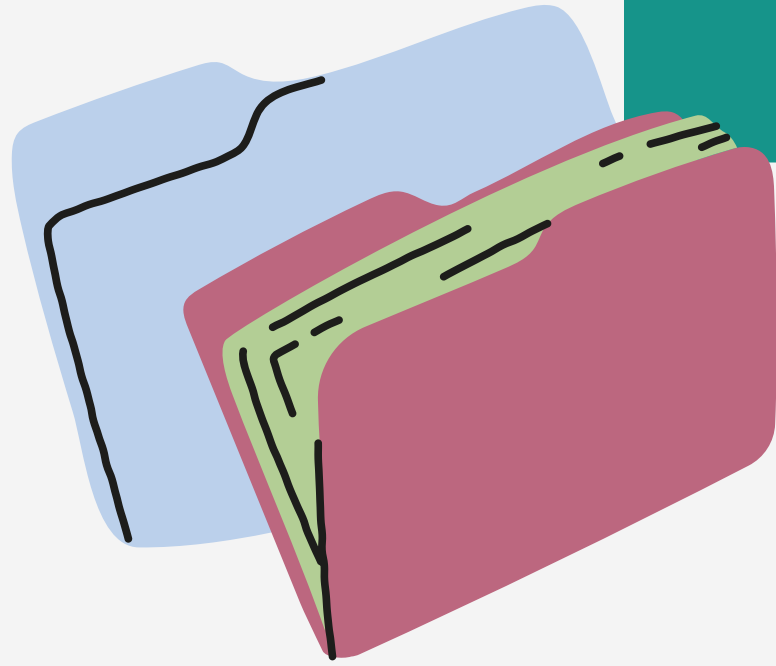


Phase 04
Speed Analysis



PHASE 01

Overview of the dataset



Minipatches_light_20260116 :

- containing experimental folders (that represents various videos)
- That contain many files



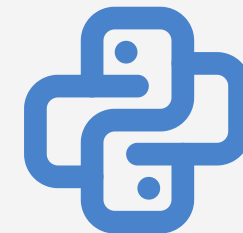
PHASE 02

Creation of a Clean Dataset



inconsistent if:

- traj.csv is missing but traj.mat is not empty.
- foodpatches_reviewed.mat is missing but traj.mat is not empty.



A Python script :

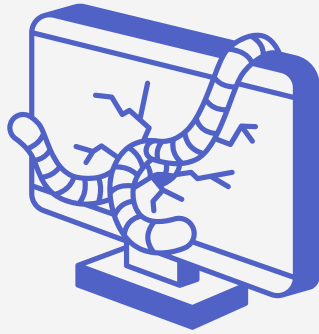
- Checks file existence.
- Measures file size -> empty files.
- Labels each video as:
OK
NO WORM
WEIRD



PHASE 03

Analysis of Multiple Worms

For each csv file:



Computation of
the proportion of
frames containing
1+ worms

Manual
inspection :
ambiguous
cases.



Assign labels:

- 0 worms
- 1 worm
- 1 worm with tracking errors
- 2+ worms



PHASE 04

Speed Analysis

For each csv file:

Computation of speed along trajectories

$$\text{Speed} = \sqrt{(xb - xa)^2 + (yb - ya)^2}$$

Plotting speed (y) over time (frames) (x)

To detect suspicious jumps => tracking errors/data corruption

Flag abnormal segments => exclusion/ correction

RESULTS

CSV files

Phase 2 : Creation of a clean dataset

Phase 3 : Analysis of multiple worms

Folder Path	traj.csv	foodpat	traj.m	traj.mat empty	Diagnosis	0 worms	1 worm	2+ worms	proportion	final classification
Documents/PTUT/	1	1	1	1	NO WORM	6255	27562	0	0.0	1 worm clean
Documents/PTUT/	1	1	1	1	NO WORM	1588	33626	0	0.0	1 worm clean
Documents/PTUT/	1	1	1	1	NO WORM	6662	4872	0	0.0	1 worm clean
Documents/PTUT/	1	1	1	1	NO WORM	3991	31033	0	0.0	1 worm clean
Documents/PTUT/	0	0	1	1	NO WORM	0	0	0	0.0	0 worms
Documents/PTUT/	1	0	1	1	NO WORM	293	1153	33511	0.9586348	2+ worms
Documents/PTUT/	1	1	1	1	NO WORM	12135	22887	0	0.0	1 worm clean
Documents/PTUT/	1	1	1	1	NO WORM	6159	28948	0	0.0	1 worm clean
Documents/PTUT/	1	1	1	1	NO WORM	8919	25983	0	0.0	1 worm clean
Documents/PTUT/	1	1	1	1	NO WORM	232	34680	0	0.0	1 worm clean
Documents/PTUT/	1	1	1	1	NO WORM	1472	33629	0	0.0	1 worm clean
Documents/PTUT/	1	0	1	1	NO WORM	622	7732	26822	0.7625085	2+ worms
Documents/PTUT/	1	1	1	1	NO WORM	3865	31358	0	0.0	1 worm clean
Documents/PTUT/	1	1	1	1	NO WORM	323	34698	0	0.0	1 worm clean
Documents/PTUT/	1	1	1	1	NO WORM	5966	29008	0	0.0	1 worm clean
Documents/PTUT/	1	0	1	1	NO WORM	16998	8147	139767756	13.9767756	1 worm with errors
Documents/PTUT/	1	1	1	1	NO WORM	13923	20675	0	0.0	1 worm clean
Documents/PTUT/	0	0	1	1	NO WORM	0	0	0	0.0	0 worms

Histogram

Phase 4: Speed Analysis

